FRISBEE

Functional Realtime Intelligence for Spotting Blissful Everyday Actions

Personal Behavioral Alignment Modeling
Using Logistic Regression on Free-Form Text

College Project Presentation

"Can a model learn who you actually are better than you know yourself?"

Abstract

FRISBEE is a personal behavioral modeling system that trains a logistic regression model on an individual's daily free-form journal messages.

The model uses four handcrafted semantic clusters — `drift`, `align`, `excuse`, `energy` — as TF-IDF weighted features $x \in \mathbb{R}^4$ to predict weekly behavioral alignment (*yes-week* vs. *no-week*) as a binary classification task.

Core Hypothesis

A model trained exclusively on the user's own language outperforms the user's own blinded self-prediction of weekly goal alignment, measured by MAE over a held-out test period.

Technical configuration: $\lambda = 0.01$ (L2, weights only) · $\eta = 0.1$ (learning rate) · Blended IDF (personal + background) · On-device computation via IndexedDB.

Introduction: The Identity Gap

There is a persistent, measurable gap between who people imagine themselves to be and who actually shows up — day to day.

Optimism Bias

Morning-you makes promises. Evening-you delivers (or doesn't). The gap is predictable.

Not a Tracker

Dashboards do not help. Understanding your personal patterns does.

The ML Question

Can a model learn *your* specific drift patterns better than you predict yourself?

Teenagers experience this most acutely:
"I said I'd study. I didn't." *"Why did I do that again?"*

This is not a productivity problem. It is a self-knowledge problem.

Objective & Scope

Objective

- Build a personal logistic regression model trained solely on one individual's daily messages.
- Predict binary weekly alignment (*yes-week* / *no-week*) from 4-cluster TF-IDF features.
- Show the model's predictions converge toward ground truth faster than the user's own blinded self-predictions.
- Demonstrate individual-level optimism bias modeling — personalized, not population-averaged.

In Scope

- Single-user, on-device computation (no server, no cloud ML)
- Free-form daily text input (no structured forms)
- Weekly binary outcome label (user-provided ground truth)
- 4 semantic clusters as fixed feature space
- Blended IDF cold-start handling

Out of Scope

Multi-user models · Automated labeling · Real-time prediction

Literature Survey

- **Sweller — Cognitive Load Theory**
  When mental capacity is depleted, people default to low-effort shortcuts; poor decision quality follows.
- **Kahneman (2011) — Thinking Fast & Slow**
  System 1 dominates when System 2 is weakened by fatigue, producing habitual poor choices.
- **Rebetez (2015) — Procrastination & Finance**
  Procrastination routes attention to immediate comfort over future consequences, systematically delaying corrective action.
- **Chwolka & Rott (2020) — Decision Deferral**
  Students overwhelmed by choices delay decisions; deferral compounds pressure and further degrades decision quality.
- **Lee & See (2004) — Trust in Automation**
  Over-trust leads to blind compliance; under-trust causes useful help to be ignored. Both extremes degrade outcomes equally.
- **Hengstler (2020) — Adaptive Trust Calibration**
  Calibrate trust to match actual system reliability over time — the

Existing Systems

- **Habit trackers** (Streaks, Habitica)
  Track completion counts — no language analysis, no prediction, no personalization of behavioral signal.
- **Journaling apps** (Day One, Reflectly)
  Store text for recall but run no modeling. Any 'insight' is surface sentiment, not behavioral prediction.
- **Population NLP** (sentiment APIs)
  Models trained on crowd data and applied to you. Your patterns are noise, not signal; individual drift is averaged away.
- **Generic recommenders**
  Suggest behavioral changes with no model of the personal gap between intention and action.

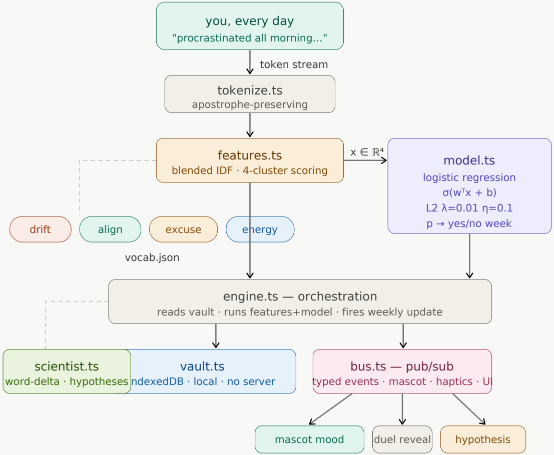
Common Gap

All existing systems are built for populations, not persons — and none model the intention-to-action gap from the user's own language.

Proposed Methodology

1. **Daily Input** User writes a free-form message about how the day went.
2. **Tokenize** Apostrophe-preserving tokenizer; contractions stay intact.
3. **Feature Extract** 4-cluster TF-IDF scoring $\rightarrow x \in \mathbb{R}^4$ vector.
4. **Predict** $z = \mathrm{w}^\top x + b \ \rightarrow \ p = \sigma(z) \ \rightarrow$ probability of *yes-week*.
5. **Weekly Label** User confirms: was this a *yes-week* or *no-week*? $\rightarrow y \in \{0, 1\}$
6. **Update** $\mathrm{w} \leftarrow \mathrm{w} - 0.1 \cdot \left[(p - y) \cdot x + 0.01 \cdot \mathrm{w} \right]$ (L2 on w only)

Detailed Architecture



Algorithm Explanation

Step 1 — Feature Engineering

Cluster	Example words
`drift`	procrastinated, wasted, scrolled
`align`	focused, proud, finished
`excuse`	tired, tomorrow, busy, forgot
`energy`	motivated, ready, pumped

TF-IDF: rare-but-used words carry more signal than frequent ones.

Blended IDF
$= 0.4 \times \text{personal} + 0.6 \times \text{background}$
(cold-start safe)

Step 2 — Model

Feature vector:
$x = [x_{\text{drift}}, x_{\text{align}}, x_{\text{excuse}}, x_{\text{energy}}] \in \mathbb{R}^4$

Linear combination:
$z = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b$

Sigmoid: $p = \dfrac{1}{1 + e^{-z}}$

Loss: $\mathcal{L} = -\big[y \log p + (1 - y)\log(1 - p)\big]$

Weight update:
$w \leftarrow w - \eta\big[(p - y) \cdot x + \lambda w\big]$

Bias update: $b \leftarrow b - \eta(p - y)$

Hyperparameters: $\eta = 0.1$, $\lambda = 0.01$, L2 on w only

Performance Metrics

MAE — Mean Absolute Error

Primary metric. Measures average prediction error vs. ground truth across weeks. Model MAE is compared to user self-prediction MAE.

Accuracy — Binary Classification

Fraction of weeks where the model's binary prediction ($p > 0.5$) matches the user's label $y \in \{0, 1\}$.

Calibration — Confidence vs. Reality

Is the model's $p = 0.7$ actually correct $\sim$70% of the time? Measured via reliability diagram over held-out weeks.

Convergence — Weight Trajectory

How many weeks until model weights stabilize? Measured by $\|\Delta w\|$ per update cycle.

Held-out test period = 4 weeks. Small sample — results are proof-of-concept, not statistically generalisable.

Constraints

- **Sample Size** 4-week held-out test $= 4$ data points. Statistical claims are constrained to proof-of-concept; no significance testing possible at this scale.
- **Cold Start** First 2–3 weeks produce near-random predictions. Blended IDF partially mitigates this but does not eliminate it.
- **Feature Space** Fixed 4-cluster vocabulary may miss user-specific language patterns outside the hand-tuned word lists. No OOV expansion.
- **Self-Report Bias** Ground truth y is user-provided. Recall bias and mood-at-labeling-time introduce noise into the training signal.
- **Single User** Model is non-transferable. Weights from one person carry no information about another person's drift patterns.
- **Trust Calibration** Per Lee & See (2004) and Hengstler (2020): early model unreliability risks under-trust; overconfident predictions risk over-reliance.

Future Enhancements

- **Multi-week Test** Extend held-out period beyond 4 weeks to enable statistically meaningful MAE comparison claims.
- **Dynamic Vocabulary** Allow the model to discover user-specific signal words outside the fixed 4 clusters (online feature selection).
- **Temporal Decay** Weight recent messages more heavily — behavioral patterns shift over months; old data should have diminishing influence.
- **Confidence Display** Show the model's uncertainty to the user (per Hengstler 2020) to support adaptive trust calibration, not blind reliance.
- **Multi-user Study** Run FRISBEE across N participants, aggregate MAE results, and enable between-person comparison of optimism bias magnitude.
- **Cognitive Load Signal** Detect Sweller-style cognitive overload from message length/complexity drops — predict bad-decision risk before the week ends.

Conclusion

- **Personal language is predictive.**

  Your daily words encode your behavioral alignment signal: drift, alignment, excuses, energy.

- **Individual $>$ population.**

  A model trained only on your data outperforms one trained on anyone else's.

- **Optimism bias is learnable.**

  The gap between morning-you and evening-you is consistent enough to model.

$\mathbb{R}^4$ feature space $\cdot$ $\lambda = 0.01$ L2 regularization $\cdot$ $\eta = 0.1$ learning rate